

2023 AMC 10B Problem 21 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 21. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 21

Problem:

Each of 2023 balls is randomly placed into one of 3 bins. Which of the following is closest to the probability that each of the bins will contain an odd number of balls?

Answer Choices:

A. $\frac{2}{3}$

B. $\frac{3}{10}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

E. $\frac{1}{4}$

Solution:

When an odd number of balls are dropped into three bins, the parities of the bins' contents are either odd-odd-odd or even-even-odd (in some order). Consider dropping the balls one at a time. Let $P(n)$ be the probability that each of the bins contains an odd number of balls (state odd-odd-odd) after $2n + 1$ balls have been deposited. Then $P(0) = 0$, and $P(n + 1)$ can be computed from $P(n)$ by considering what happens with the next 2 balls. With probability $P(n)$ the state is currently odd-odd-odd, and there are 9 equally likely ways for 2 more balls to land. With probability $\frac{3}{9} = \frac{1}{3}$ they will land

in the same bin and the new state will be odd-odd-odd. With probability $1 - P(n)$ the state is currently even-even-odd, and 2 of the 9 possibilities for the next 2 balls will change the state to odd-odd-odd. Therefore

$$P(n+1) = P(n) \cdot \frac{1}{3} + (1 - P(n)) \cdot \frac{2}{9},$$

which simplifies to

$$P(n+1) = \frac{1}{9} \cdot P(n) + \frac{2}{9}.$$

This linear recurrence can be solved by assuming that $P(n) = a \cdot r^n + b$ for some constants r , a , and b . Because $P(0) = 0$, it follows that $a \cdot r^0 + b = 0$. Because

$$P(1) = \frac{1}{9} \cdot P(0) + \frac{2}{9} = \frac{1}{9} \cdot 0 + \frac{2}{9} = \frac{2}{9},$$

it follows that $a \cdot r^1 + b = \frac{2}{9}$. Because

$$P(2) = \frac{1}{9} \cdot P(1) + \frac{2}{9} = \frac{1}{9} \cdot \frac{2}{9} + \frac{2}{9} = \frac{20}{81},$$

it follows that $a \cdot r^2 + b = \frac{20}{81}$. Solving this system of equations gives $r = \frac{1}{9}$, $a = -\frac{1}{4}$, and $b = \frac{1}{4}$. Therefore

$$P(n) = -\frac{1}{4} \cdot \left(\frac{1}{9}\right)^n + \frac{1}{4}$$

The recurrence and initial condition are satisfied by this formula for $P(n)$, so the assumption about the form of $P(n)$ was justified. The problem asks for $P(1011)$, which nearly equals (E) $\boxed{\frac{1}{4}}$ because the first term is very close to 0.

OR

Let O and E represent parities odd and even. Because the number of balls is odd, there are only 4 possible parity states of the three bins:

EEO, EOE, OEE , and OOO . The problem asks for

the approximate probability that the state is OOO . If the number of balls is sufficiently large, the probability that the first bin has an even number of balls is very nearly $\frac{1}{2}$. The same holds for the second bin. Thus EE, EO, OE , and OO are all approximately equally likely for the parities of the first two bins.

The parity of the third bin is completely determined by the parities of the first two bins because the total number of balls is odd. Thus the probability of

each of the 4 parity states is about $\frac{1}{2} \cdot \frac{1}{2} = \left(E\right) \boxed{\frac{1}{4}}$.

OR

If the balls are placed randomly into the bins one at a time, after n balls are placed, either

- all three bins contain the same parity of balls (that is, either all three bins contain an odd number of balls or all three bins contain an even number of balls),
or
- two of the bins contain one parity of balls and the third bin contains the other (that is, either two bins have an an odd number of balls and the remaining bin has an even number of balls, or two bins contain an even number of balls and remaining bin has an odd number of balls).

Note that if all three bins currently contain the same parity of balls, they will surely not contain the same parity once another ball is added to any bin.

However, if two bins currently contain the same parity of balls and the third bin contains the other parity, a ball added to the third bin will make all three bins contain the same parity, but one added to either of the first two bins will result in two of the bins having the same parity but not the remaining bin.

Let p_n represent the probability that all three bins contain the same parity of balls. From the reasoning above, $p_{n+1} = 0 \cdot p_n + \frac{1}{3} (1 - p_n) = \frac{1-p_n}{3}$. Assuming that p_n approaches a limit L as $n \rightarrow \infty$, the recurrence implies that $L = \frac{1-L}{3}$, from which it follows that $L = \frac{1}{4}$. Therefore p_{2023} should be

close to $\left(E\right) \boxed{\frac{1}{4}}$.

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